

ARTICLE TYPE

Deep-Reorder: Refined Motion Correction for 3D Brain MRI via Iterative Data Regrouping

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Summary

Purpose: Accurate and temporally high-resolution estimation of subject motion is paramount for effective motion-correction in MRI. This work proposes a novel approach for estimating subject motion within a 3D MRI scan with high temporal resolution.

Methods: Starting from an incoherent sampling scheme, images are reconstructed from temporally ordered k-space fractions via deep learning, enabling motion estimation via registration between each image. Assuming the bulk of the motion is removed using these initial estimates, the remaining misalignment between k-space fractions is expected to be minimally correlated in time. Based on this insight, an iterative strategy is proposed in which shots are arbitrarily regrouped into new k-space fractions, regardless of their original temporal ordering, for subsequent motion estimation and correction. This effectively increases the temporal resolution of the approach, enabling individual correction of all shots.

Results: When applied to 3D T1w TFE MRI of the brain, the proposed approach “Deep-Reorder” estimated a unique motion state for all shots, producing significantly better estimations of retrospective motion compared to estimation without iterative data regrouping. On prospectively motion-corrupted scans with intentional head movement by volunteers, the proposed approach consistently improved image quality, producing less noisy images with reduced motion artifacts.

Conclusion: In conclusion, the proposed approach, which is free of external tracking devices or navigators, effectively estimated 3D motion and corrected motion artifacts with increased temporal resolution.

KEYWORDS:

Retrospective Motion Correction, MRI Motion Artifacts, Deep Learning Reconstruction, Motion Estimation, Head Motion, Temporal Resolution

1 | INTRODUCTION

Patient motion during MRI acquisition can result in artifacts typically seen as ghosting and blurring that degrade diagnostic quality and may in severe cases necessitate a repeat scan. This disproportionately affects populations who find it difficult to remain still, such as children, elderly patients, and those with movement disorders. The effects of motion have measurable financial and throughput consequences for imaging centers, and estimates of annual per-scanner costs attributed to motion can run into the hundreds of thousands of dollars^{1,2}. Consequently, methods that mitigate motion artifacts can partially relieve these problems.

Retrospective motion-correction operates after data collection, seeking to remove or reduce motion-induced artifacts during the reconstruction stage. In recent years, end-to-end AI-driven reconstruction frameworks that learn mappings from motion-corrupted to motion-free images have gained traction due to rapid inference, relatively straightforward implementation, and the increasing availability of training data. These models are often trained on synthetic datasets in which motion is introduced to motion-free images through physics-based simulations of k-space corruption, allowing the network to learn to reduce motion artifacts on paired data in a supervised manner^{3,4,5}. Recent work has expanded beyond supervised regression models toward generative approaches⁶, particularly diffusion models^{7,8}, which have shown promise in reconstructing high-fidelity images by iteratively denoising motion-corrupted images. While such methods can produce visually convincing results, their lack of an explicit motion model means that they must deviate from the acquired k-space data to remove artifacts. Under substantial motion, the reliance of these methods on learned priors increases the risk of hallucinations, namely omitting true anatomy or introducing features that are not present in the underlying data^{9,7,10}.

Data-consistent retrospective motion correction approaches are preferred for this reason, but they must solve an ill-posed inverse problem that couples image content and motion, generally requiring a parameterized estimation of the motion.

External tracking hardware can supply motion parameters but often adds complexity to the workflow, raises cost, and may not always deliver sufficiently accurate measurements. Data-driven correction strategies that infer motion directly from the imaging data are attractive as they avoid extra equipment and integrate more naturally into clinical protocols¹¹.

Many approaches jointly search for the motion parameters and motion-corrected image by alternately

solving for one given the other^{12,10,13,14}. Notable among these techniques is the DISORDER framework¹⁵, which incorporates a dedicated k-space sampling strategy designed to make the motion easier to detect by incoherently distributing low and high frequencies across all shots.

In order to accurately describe the motion, one needs to consider sufficiently small time intervals that are at the time scale of the motion or smaller. However, the shorter the intervals, the less data is available per interval and the more difficult it is to estimate the motion parameters. For example, it is not feasible to infer the subject's orientation from a single line in k-space. Naturally, attempting to estimate motion from too short time frames can lead to inaccurate estimations that would even deteriorate the motion-corrected reconstruction. Existing approaches therefore often opt to estimate motion over longer time intervals, guaranteeing that each interval contains enough data to reliably estimate the motion state, though the effective temporal resolution of the motion estimation is then per definition compromised, leaving some motion artifacts.

We propose a novel approach, Deep-Reorder, for estimating subject motion with increased temporal resolution. We demonstrate its effectiveness by combining this idea with DEEP-DISORDER, our previously proposed motion estimation technique that can reliably estimate motion for k-space fractions of only ~2.3% of the data. Rather than using even smaller k-space fractions, which may contain too little information and result in inaccurate estimations, the proposed approach iteratively recombines data to generate new k-space fractions and corrects the residual motion therein. This effectively increases the temporal resolution to the shot level, at ~0.8% of the data for the investigated 3D T1w TFE sequence.

2 | METHODS

2.1 | Problem Definition

The acquisition of k-space y is described by the forward model $y = Ax$, with anatomical image x and the forward operator $A = MFSC_\theta$. The C_θ operator describes motion-corruption of x in a shotwise manner by projecting the image into different motion states, using rigid motion parameters θ with 3 translations and 3 rotations per shot. S denotes the coil-sensitivity encoding operator, which pointwise multiplies the motion-transformed image of each shot by the static coil sensitivity maps, so

that motion affects which anatomy is sensed by each coil.¹⁴⁸
 F represents the Fourier transform and M denotes the¹⁴⁹
masking operator that selects the k-space lines measured¹⁵⁰
in each shot. Estimated motion parameters $\hat{\theta}$ can be used¹⁵¹
to reverse displacements and rotations that occurred dur-¹⁵²
ing acquisition on a per-shot basis. Specifically, we use¹⁵³
CG-SENSE adapted for motion compensation^{16,17,18,19}¹⁵⁴
to solve the motion-corrected reconstruction problem¹⁵⁵

$$\hat{x} = \underset{x}{\operatorname{argmin}} \|MFSC_{\hat{\theta}}x - y\|_2^2. \quad (1)$$

The main challenge lies in estimating the motion param-¹⁵⁶
eters $\hat{\theta}$ from the measured k-space data y .¹⁵⁷
¹⁵⁸
¹⁵⁹
¹⁶⁰
¹⁶¹
¹⁶²

2.2 | DEEP-DISORDER

In a previous work, we developed an approach that recon-¹⁶³
structs fractions of k-space and registers them to esti-¹⁶⁴
mate rigid motion parameters for each²⁰. This approach,¹⁶⁵
termed DEEP-DISORDER, can be summarized as fol-¹⁶⁶
lows. A 3D-Unet²¹ was trained for this reconstruction¹⁶⁷
task, which produces low-quality images due to the¹⁶⁸
short time window. The images are nonetheless accu-¹⁶⁹
rate enough to estimate the position of the subject at¹⁷⁰
that time using groupwise registration, for example with¹⁷¹
Elastix^{22,23,24}. Depending on the sequence, there can be
too little information in a single shot for reliable motion
estimation, in which case multiple shots are grouped into
a “segment” for which a joint motion state is estimated.
Specifically, in the case of 3D T1-weighted TFE for which
133 shots are acquired, 3 shots are grouped into one
segment. Figure 1 b illustrates the DEEP-DISORDER
approach. The method uses incoherent DISORDER sam-
pling to ensure a mix of low and high frequencies in
every shot, making the motion easier to estimate¹⁵. In
the previous work, this approach was demonstrated to
have similar performance to the aligned reconstruction
of DISORDER¹⁵, a state of the art motion estimation
and correction algorithm.

2.3 | Iterative Regrouping

With DEEP-DISORDER, motion that occurs within the
segments is uncorrected. Ideally, a motion state would
be estimated for each shot, but several shots need to
be grouped into one segment to have enough informa-
tion for accurate motion estimation²⁰. The conventional
algorithm makes use of temporal adjacency in the mea-
sured data through contiguous segments, by assuming
that shots which are adjacent in time are likely to be the
closest in motion state and, therefore, can be grouped
together. We postulate that initial motion-correction suf-
ficiently aligns temporally non-adjacent segments, such

that any shot can hereafter be regrouped with any other
shot. Thus, after initial correction, we propose to con-
tinually regroup the shots into different segments, and
estimate residual motion parameters for these precor-
rected segments, which can be used to further refine the
previous motion parameters on a per-shot basis. While
the number of motion states estimated during a single
iteration is still limited to the number of segments,
after multiple iterations there will be a unique esti-
mated motion state for each of the shots. This concept
is visualized in Figure 2. Figure 1 a illustrates the pro-
posed iterative regrouping approach, while Algorithm 1
describes the same concept algorithmically. Executing
one iteration of this algorithm corresponds to the existing
DEEP-DISORDER approach.

Naturally, the residual motion can be reduced most
quickly when shots are grouped that are similar in terms
of motion state. Determining which shots are the closest
in terms of motion state is by no means trivial, as the
premise of our problem is estimating the motion states
in the first place. We use a random segment regrouping
strategy, which, compared to a heuristic selection of seg-
ments, is algorithmically straightforward, may generalize
well, and is computationally inexpensive.

Algorithm 1 Proposed Iterative Regrouping Algorithm

Data: Motion-corrupted k-space y , forward operator A ,
segment reconstruction function f_{recon}^G using grouping G
defining which segment each shot belongs to, registra-
tion function f_{reg} , n segments and m shots with $n < m$,
rearrangement function ρ , motion-corrected reconstruc-
tion algorithm $\mathcal{R}$.

Result: Corrected image $\hat{x}$, estimated motion param-
eters $\hat{\theta}$.

- 1: $\hat{\theta} \leftarrow 0$ *Initialize estimated motion parameters.*
 - 2: $G \leftarrow [1, 1, \dots, 2, 2, \dots, n, n, \dots]$ *First few shots*
are in segment 1, next few are in segment 2, etc.
 - 3: **while** stopping criterion not reached **do**
 - 4: $\bar{x} = A_{\hat{\theta}}^H y$ *Generate pre-corrected image*
using estimated motion $\hat{\theta}$.
 - 5: $\hat{\theta} \leftarrow \hat{\theta} + f_{\text{reg}}(f_{\text{recon}}^G(\bar{x}))$ *Estimate incremental*
motion update by reconstructing
and registering segments defined by G .
 - 6: $G \leftarrow \rho(G)$ *Regroup shots into random segments.*
 - 7: **end while**
 - 8: $\hat{x} = \mathcal{R}(y, \hat{\theta})$ *Final reconstruction using*
estimated motion.
 - 9: **return** $\hat{x}, \hat{\theta}$
-

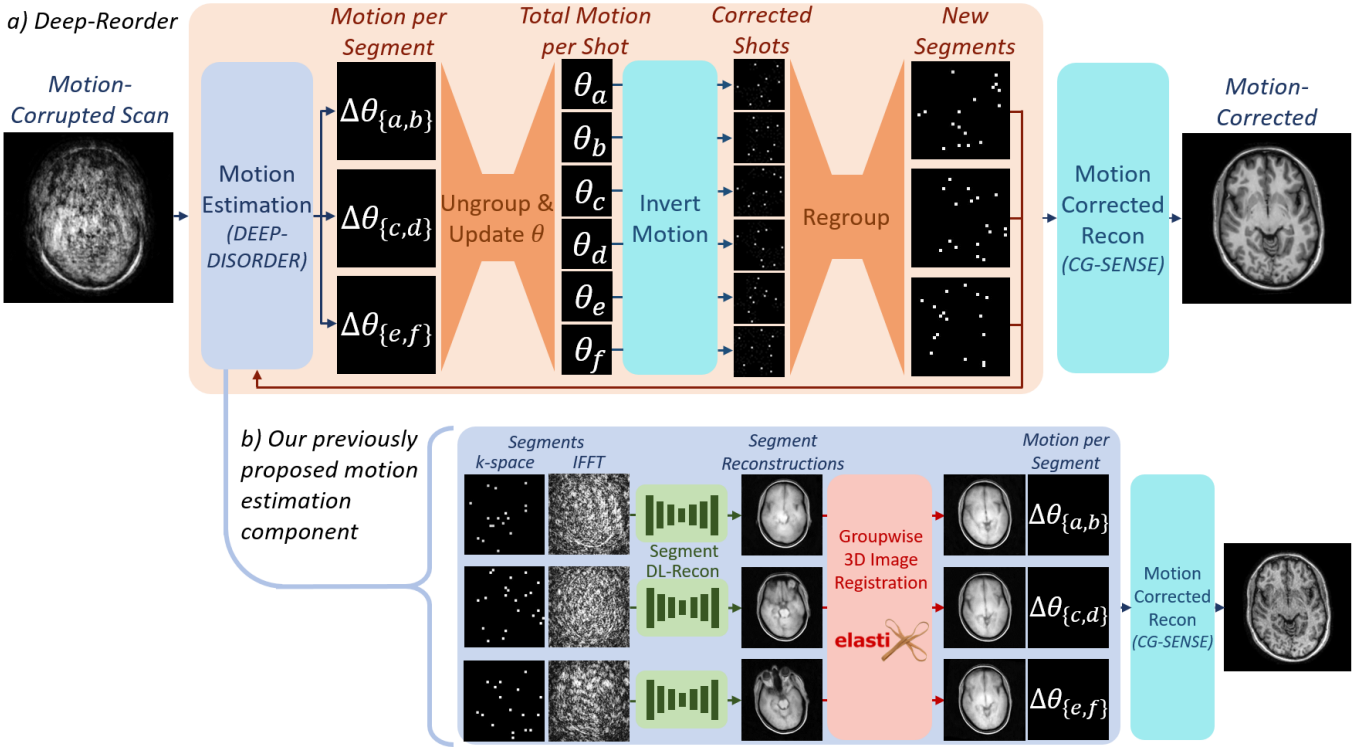


FIGURE 1 a: The proposed Deep-Reorder approach iteratively re-uses an existing motion estimation method. After motion is estimated for a segment, which consists of multiple shots (e.g. shots a and b), the individual motion parameters of those shots are updated accordingly. Next, the total parameters are used to correct the motion in the measured data, and new segments are made from different combinations of shots, for which more residual motion may be removed in the next iteration. In the end, each shot can be corrected with its own motion parameters in a final motion-corrected reconstruction. **b:** The existing approach DEEP-DISORDER estimates motion on a per-segment basis.

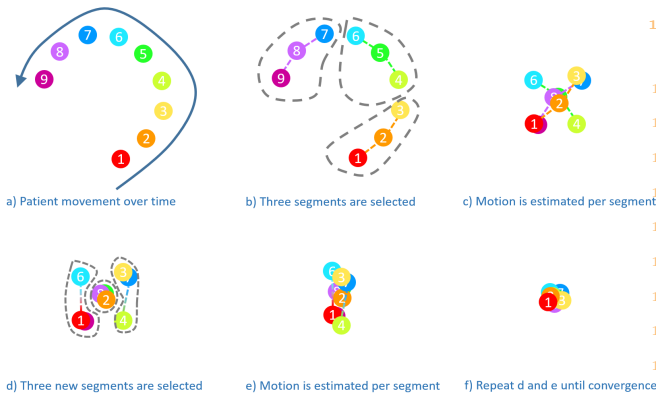


FIGURE 2 Given a simplified motion-corrupted scan with 9 motion states (a), an existing motion correction algorithm accurately estimates motion for 3 segments, but cannot solve intrasegment motion (b, c). The proposed iterative regrouping algorithm improves the temporal resolution of the existing motion correction algorithm from 3 to 9 estimated motion states (d, e, f).

2.4 | Maximum Motion Update

Rather than directly adding the newly estimated residual motion parameters, we limit updates of the motion to at most $\Delta_{\max}^k$ via vector-norm clamping, to reduce the impact of potential outlying incorrect estimates and to reduce potential fluctuations when the remaining motion is as small as the expected intersegment error of the used motion estimator. This limit is initially high to enable large changes when the motion residuals are still high, and then decreases according to

$$\Delta_{\max}^k = c/(k + 1/\mu), \quad (2)$$

where k is the current iteration. During the first iteration, which is equivalent to conventional correction, large intersegment motion needs to be corrected and therefore no maximum motion limit is used. Based on this initial estimation of intersegment motion, $\mu = \sum_j \sum_i |\theta_{i+1,j} - \theta_{i,j}|$ quantifies the estimated motion severity given segment index i and six degrees of freedom j . Thus, larger initial estimated motion leads to a higher maximum motion update limit. The underlying rationale is that a large initial motion estimate is indicative of large intrasegment

motion, in which case the iterative algorithm benefits from a larger maximum motion update for faster convergence. Conversely, if the initial estimated motion is low, sudden large residual motion estimations are more likely to be mispredictions and are clamped more strongly. We selected $c = 10$. For two hypothetical examples of $\mu = 0.05$ – 0.5 , this yields maximum translations and rotations $\Delta_{\max}^k$ of 0.4 – 1.4 mm and degrees at iteration $k = 5$ and 0.2 – 0.4 mm and degrees at iteration $k = 25$.

As a stopping criterion for our algorithm, we use the reference-less Laplacian variance sharpness metric. This metric evaluates image focus by computing the variance of the Laplacian response, which increases with high-frequency detail²⁵. If the latest solution has not improved over the best solution in terms of Laplacian variance for a defined number of iterations, the algorithm is stopped and the best solution is accepted. In other words, given the Laplacian variance metric V , stop at iteration k and accept the best corrected image $\hat{x}_{\text{best}}$ if

$$\forall t \in \{k-9, \dots, k\} : V(\hat{x}_t) \leq V(\hat{x}_{\text{best}}). \quad (3)$$

where

$$\hat{x}_{\text{best}} = \arg \max_{0 \leq s \leq k} V(\hat{x}_s). \quad (4)$$

2.5 | Training and Implementation Details

The segment reconstructions of DEEP-DISORDER are performed by an unrolled 3D U-Net^{26,21,27} with 16, 32, 64, and 128 feature maps per level. The model uses group normalization, 3D convolution and ReLU operators in each layer. An equally weighted sum of SSIM and smooth-L1 loss was used during training, which was calculated between a motion-free fully sampled 3D image as target and the estimated segment reconstruction. The latter was estimated by the network based on the highly undersampled zero-filled 3D image that it received as input. To account for the various positions the subject may be in, the training data was randomly translated and rotated. Two segment reconstruction models with identical architecture were trained, one for segments of acceleration factor $R = 1$ and one for $R = 2$ as shots from accelerated scans sample more densely from the k-space center. Training was performed on an NVIDIA RTX A6000 GPU for 3×10^5 iterations, using a batch size of 2 and a group normalization size of 8. Finally, accounting for the fact that the volunteer scans outlined in Section 2.7 may be out-of-distribution, the models were finetuned using leave-one-out cross-validation on motion-free volunteer scans. Specifically, starting from the pretrained model, the training was continued using

the same settings for 1×10^5 iterations on an equal proportion of segments from volunteer scans and the original training scans. No layers were frozen during fine-tuning, and the same optimizer and learning rate were used as during the initial training. An overview of the number of finetuned models is given in Table 1. Apart from using different finetuned segment reconstruction models, the approach uses the same hyperparameters for the various datasets.

Groupwise registration was performed using the Elastix registration toolbox. A multi-resolution pyramid of 4 resolutions was used, and was run for 2500 iterations per resolution. The objective function of the groupwise registration to find rigid motion parameters $\{\hat{\theta}_1, \dots, \hat{\theta}_N\}$ is described by

$$\{\hat{\theta}, \hat{T}\} = \underset{\theta, T}{\operatorname{argmin}} \sum_{s=1}^N \left\{ \sum_v \left\| \theta_s(\hat{x}_s(v)) - \bar{x}(v) \right\|_2^2 + \operatorname{dist}(e, \theta_s) \right\}. \quad (5)$$

Here, θ_s is the transformation for segment s , $\hat{x}_s(v)$ is the reconstructed image for segment s and voxel v , $\bar{x}(v)$ is the common groupwise template which is computed as the average of the transformed images and updated iteratively, and $\operatorname{dist}(e, \theta_s)$ is a regularization term enforcing closeness to the identity transformation e ²⁸. Trilinear interpolation was used in the motion-corruption operator C_{θ} .

2.6 | Evaluation Metrics

To assess the quality of the reconstructed images, we employ two widely used reference-based image quality metrics: Structural Similarity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR)^{29,30}. SSIM evaluates perceptual similarity by comparing luminance, contrast and structural information between the reconstructed image and the ground truth. PSNR quantifies the pixelwise fidelity between the reconstructed image and the ground truth. The ground truth is either the uncorrupted image in the case of retrospective motion, or a separately acquired motion-free reference scan in the case of prospective motion.

2.7 | Data

Retrospectively motion-corrupted data: An in-house collection of 403 volumetric brain scans without intentional motion was used, acquired with a 3D T1-weighted turbo field echo (TFE) sequence. All participants provided written informed consent and the study protocol was approved by the institutional review board. The dataset was partitioned into a training and validation set of 361 cases and a test set of 42 cases. All scans

Dataset	Training	Evaluation
Retrospectively Motion-Corrupted	Training on motion-free patient scans (1 model for $R = 1$, 1 model for $R = 2$)	Nodding, Perlin Random, Uniform Random, Normal Random $R = 1$
Prospectively Motion-Corrupted	Leave-one-out finetuning on motion-free volunteer scans (3 models for $R = 1$, 3 models for $R = 2$)	Gradual, Jittery, Sweeping $R = 1$; Gradual, Jittery, Sweeping $R = 2$

TABLE 1 Overview of the number of trained and finetuned segment reconstruction models, as well as an overview of the motion-corrupted datasets used for evaluation, where R stands for acceleration factor.

were acquired on 3T scanners (Ingenia 3.0T and Ingenia Elition 3.0T X, Philips Healthcare, Best, The Netherlands). The sequence parameters included a flip angle of 8° , repetition time $TR \approx 9.9$ ms, and echo time $TE \approx 4.6$ ms. Data were collected with an effective acceleration of $R = 1.8$. Coil-combined images were obtained using the vendor’s compressed sensing reconstruction software. As the original acquisition matrices varied across subjects, each volume was resampled by cropping or zero-padding in k-space to a uniform size of 192^3 voxels. Coil sensitivity maps were extracted from the SENSE reference scan that is automatically acquired with sequences requiring sensitivity information. Sensitivity maps were compressed to 8 virtual coils using singular value decomposition-based array compression and were normalized using root-sum-of-squares normalization.

Paired motion-free and retrospectively motion-corrupted k-space datasets were produced by forward-simulating the MR acquisition from the motion-free coil-combined images, applying different 3D rigid motion to each of the 133 shots. The “nodding” motion pattern uses intentional head nodding and shaking trajectories captured on a scanner with an optical tracking system from a publicly available dataset^{32,33}, from which we sampled 133 motion states at uniform temporal intervals. The “Perlin random” motion pattern uses smooth trajectories generated using Perlin noise³⁴, a procedural noise function commonly used in computer graphics to generate correlated random variations. We used a base amplitude of 8 and four noise periods along each spatial axis to produce coherent low-frequency motion. The “uniform random” motion pattern uses 133 motion states sampled independently along each axis from a uniform distribution with $[-4, 4]$ mm and degrees per axis, while the “normal random” pattern is sampled from a normal distribution with $\sigma = 4$ mm and degrees. An example of each pattern can be seen in Figure 3. The simulation emulated the TFE acquisitions with 216 k-lines per shot, matching the TFE factor of the prospectively motion-corrupted data.

Prospectively motion-corrupted data: Additional 3D T1-weighted prospectively motion-corrupted datasets were obtained from our previous study²⁰, consisting of three healthy volunteers who provided written

informed consent, whose scans were used for testing. An additional independent volunteer was scanned beforehand during development of the approach and was used to establish standardized instructions, but was not included in the test set. The scans used a 3D TFE sequence on a 1.5T scanner (Ingenia, Philips Healthcare, Best, The Netherlands), using DISORDER sampling¹⁵. The sequence parameters included a flip angle of 8° , repetition time $TR \approx 4.6$ ms, and echo time $TE \approx 2.1$ ms. Volumes were acquired with a matrix size of $191 \times 191 \times 384$ and a TFE factor of 216. Coil sensitivity maps were obtained from the automatically acquired SENSE reference scan, compressed to eight virtual coils, and normalized prior to use in reconstruction. Data were collected using an elliptical shutter with prospective accelerations of $R = 1.0$ and $R = 2.0$, consisting of 133 and 68 shots, respectively. A DISORDER sampling pattern was used that samples the same lines as the conventional compressed sensing acquisitions, matching the elliptical shutter and denser sampling near the k-space center for accelerated scans. During reconstruction, we discarded the last shots to make the total number divisible by 3 and keep the algorithm simpler. This reduced the number of shots used to 132 and 66, thereby slightly increasing the undersampling factor of the data to 1.01 and 2.06. Alternatively, it would be possible to correct the remaining shots by randomly leaving different shots out per iteration or by adding a few duplicate shots, but in our case the acceleration is low enough that the CG-SENSE reconstruction can recover the discarded data. Total acquisition times were 6 min 44 s for $R = 1$ and 3 min 29 s for $R = 2$.

During each scan, volunteers were instructed to follow one of three predefined motion patterns so that characteristic forms of head motion could be reproduced under controlled conditions. In the “gradual” motion condition, volunteers occasionally performed slow head rotations to mimic the drift commonly observed in clinical examinations as subjects relax or fall asleep. In the “jittery” motion condition, volunteers were instructed to make abrupt head rotations in a random direction at intermittent intervals. In the “sweeping” motion condition, volunteers began the scan looking as far as possible toward the lower left and progressed in discrete steps

until they were facing the upper right. In some cases, the instructed motion was completed early and the volunteers reversed direction toward the starting pose. The acquisitions were performed without padding in the head coil to allow larger head excursions. Finally, motion-free scans were acquired for reference. An overview can be seen in Table 1.

3 | RESULTS

3.1 | Evaluation on Retrospectively Motion-Corrupted Data

Figure 3 shows DEEP-DISORDER and the proposed approach on retrospectively motion-corrupted 3D T1w TFE scans. As illustrated by the plots of the estimated motion parameters, DEEP-DISORDER cannot correct 132 states since it needs to group 3 shots per segment to have enough information to make an accurate estimation of the anatomy. The proposed approach is capable of making a unique motion estimation for each of the 132 states, and is consequently able to correct most of the motion artifacts. The random motion patterns in particular, while unlikely to occur in in vivo subjects, aptly demonstrate the increased temporal resolution of Deep-Reorder.

Figure 4 shows the performance of the proposed approach per iteration. For these experiments, the algorithm was run for a fixed number of iterations, regardless of the stopping criterion. It can be seen that the solution gradually improves as more iterations are performed up to a certain lower bound. Since motion estimation realistically has limited accuracy, the iterative solution will also contain non-perfectly aligned states, and therefore the error does not reach zero. Nonetheless, the final motion estimation by Deep-Reorder reached sub-voxel accuracy and is sufficiently accurate so that the resulting images appear virtually free of motion artifacts. Figure 5 illustrates the quality of the corrected image as more iterations are performed. Table 2 shows the quantitative performance of both approaches.

3.2 | Evaluation on Prospectively Motion-Corrupted Data

Table 3 compares uncorrected reconstructions, DEEP-DISORDER and the proposed Deep-Reorder approach on prospectively motion-corrupted 3D T1w TFE scans of volunteers with intentional motion. The metrics were computed after registering the reconstructed scans to corresponding fully sampled motion-free reference scans. Figure 6 displays examples of motion-corrected images.

The top row shows a case in which DEEP-DISORDER already performed well, leaving little room for improvement by Deep-Reorder. The other rows provide examples of cases in which the reconstructions still show signs of motion-corruption after correction with DEEP-DISORDER. Deep-Reorder further reduced the corruption across all such cases, yielding noticeably sharper and less noisy reconstructions. In the second and third rows of full-brain images, prominent streaking artifacts are visible that are substantially diminished by Deep-Reorder. The bottom two rows illustrate worst-case examples of corruption induced by particularly severe motion, where despite Deep-Reorder considerably improving over DEEP-DISORDER, substantial motion-related degradation remains in the final reconstruction.

3.3 | Cold-diffusion-based Refiner

In this experiment, we investigate the use of a cold-diffusion-based refiner to correct residual motion artifacts and amplified noise that cannot be completely accounted for in a motion-corrected CG-SENSE reconstruction. Cold diffusion learns to invert a structured degradation process³⁶, which in our case is retrospective motion corruption that models residual motion artifacts at each diffusion timestep, learned by a 3D-Unet²¹ with four encoder-decoder levels. Specifically, we define the motion artifacts for training to be the difference between an uncorrupted image and a synthetically lightly retrospectively motion-corrupted counterpart. The motion was sampled independently along each axis from a normal distribution with $\sigma = t \cdot 0.12$ mm and degrees for t between 0 and 10. An equally weighted sum of SSIM, Huber and Sobel loss was used during training. For data augmentation, the training data was randomly translated and rotated. The images were normalized linearly to a range of -1 to 1 , while clamping the lowest and highest percentile of voxels in terms of intensity. During inference, four reverse diffusion steps were performed. At each step, the network predicted the residual motion artifacts corresponding to the current diffusion timestep, which was subtracted from the motion-corrected reconstruction before proceeding to the next timestep, ultimately reaching $t = 0$. The results can be seen in Figure 6, where particularly in the zoomed-in regions, the refiner reduces the residual noise-like artifacts that remained after reconstruction with Deep-Reorder.

3.4 | Maximum Motion Update Study

Figure 7 investigates the scale factor c used in the maximum motion update function. It can be seen that

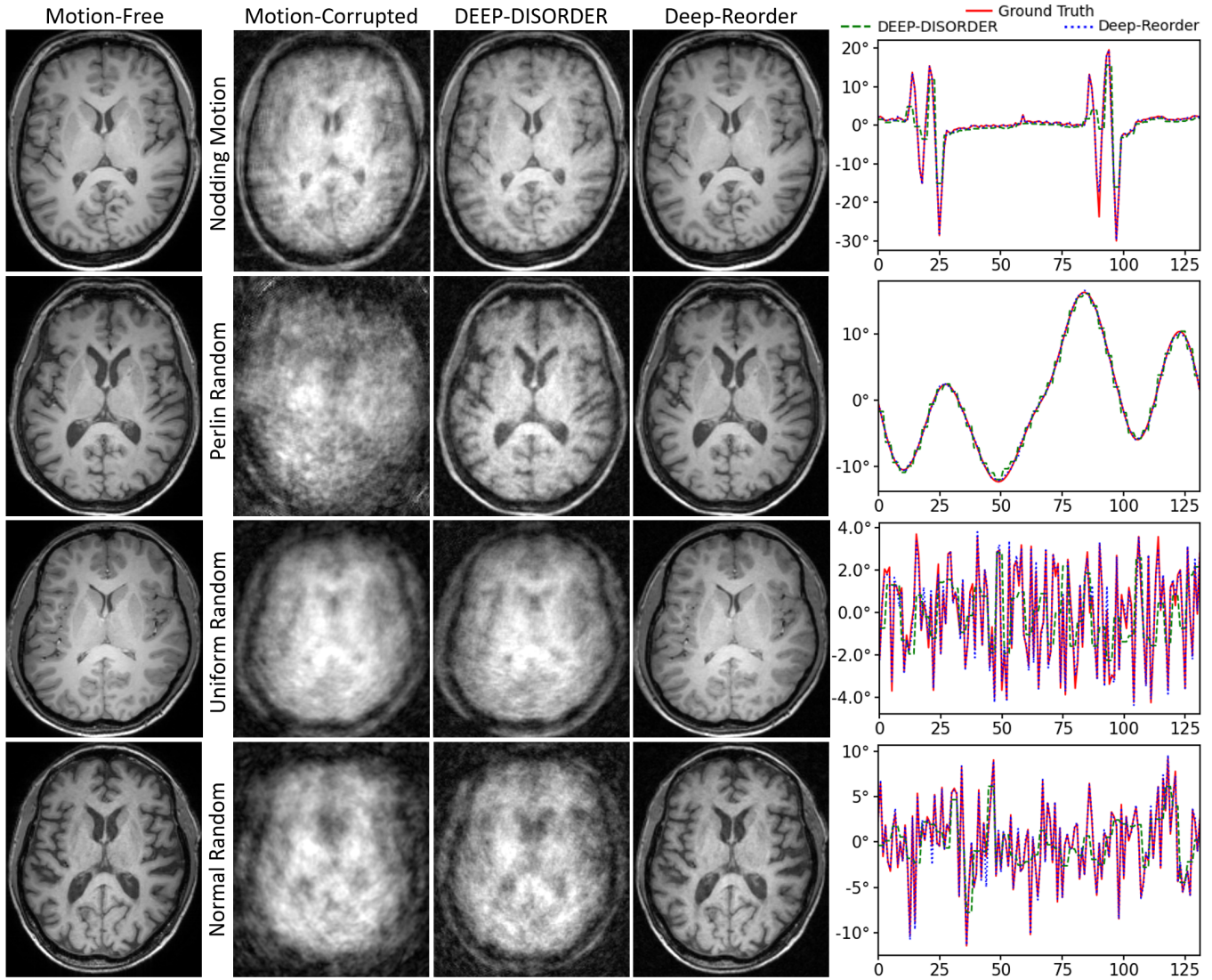


FIGURE 3 Reconstructions of retrospectively motion-corrupted data with 132 motion states. The first row displays nodding motion from a tracking device. The second row displays a pattern sampled from Perlin noise. The pattern in the third row was generated using 132 motion states sampled uniformly over $[-4, 4]$ mm or degrees per degree of freedom, while the fourth pattern was generated using a normal distribution with $\sigma = 4$. The plots show the estimated and known rotational motion about the z-axis (feet-head).

for retrospectively motion-corrupted data, the applica-
tion of the maximum motion update does not improve
the result, and may deteriorate the result when it is
applied too strongly. For prospectively corrupted data,
while a strong limit is still detrimental, a moderate limit
of around $c = 5$ to 10 is required to increase the speed
of convergence and prevent the degradation that other-
wise happens in a few volunteer scans. This may be a
consequence of the segment reconstruction model being
less accurate on the volunteer scans, while the motion
update limit reduces the impact of fluctuations in the
estimated motion parameters. The reference-less Lapla-
cian variance metric was used to guide the selection for
 c , setting it to 10 for all data, while the evaluation in

terms of SSIM was conducted post hoc to prevent the
parameter selection from being informed by the reference
scans.

3.5 | Alternative Regrouping Strategies

In addition to the proposed random regrouping strategy,
we compare two alternative strategies: *sliding window*
regrouping, which repeatedly groups temporally adjacent
shots, and *dilated* regrouping, which during the second
iteration groups shots that had the same relative posi-
tion within their original segments. These strategies are
designed to test how different grouping patterns affect
the convergence of the motion estimation approach. For

Motion Pattern	Reconstruction Approach	Motion-Corrected Reconstruction Quality ($\mu \pm \sigma$)		
		Motion Error ($^\circ$) $\downarrow$	SSIM $\uparrow$	PSNR (dB) $\uparrow$
Nodding	Uncorrected	2.59 ± 1.05	0.898 ± 0.026	28.2 ± 1.6
	DEEP-DISORDER	$0.76 \pm 0.27^*$	$0.973 \pm 0.010^*$	$35.7 \pm 2.0^*$
	Deep-Reorder	$0.27 \pm 0.09^{*\dagger}$	$0.992 \pm 0.005^{*\dagger}$	$42.1 \pm 2.5^{*\dagger}$
Perlin Random	Uncorrected	6.38 ± 0.12	0.794 ± 0.016	23.0 ± 1.0
	DEEP-DISORDER	$0.63 \pm 0.03^*$	$0.952 \pm 0.006^*$	$32.9 \pm 1.1^*$
	Deep-Reorder	$0.25 \pm 0.06^{*\dagger}$	$0.992 \pm 0.004^{*\dagger}$	$42.2 \pm 1.9^{*\dagger}$
Uniform Random	Uncorrected	1.92 ± 0.04	0.837 ± 0.014	25.8 ± 0.9
	DEEP-DISORDER	$1.56 \pm 0.03^*$	$0.865 \pm 0.013^*$	$27.2 \pm 0.9^*$
	Deep-Reorder	$0.21 \pm 0.05^{*\dagger}$	$0.994 \pm 0.003^{*\dagger}$	$43.4 \pm 2.2^{*\dagger}$
Normal Random	Uncorrected	3.30 ± 0.08	0.805 ± 0.015	23.7 ± 1.0
	DEEP-DISORDER	$2.76 \pm 0.04^*$	$0.822 \pm 0.015^*$	$24.9 \pm 0.9^*$
	Deep-Reorder	$0.25 \pm 0.06^{*\dagger}$	$0.991 \pm 0.003^{*\dagger}$	$40.6 \pm 2.5^{*\dagger}$

TABLE 2 Reconstruction quality and accuracy of the estimated motion on retrospectively motion-corrupted data. The motion error was calculated as the mean difference between the estimated and ground truth motion over all shots and 3 degrees of freedom, in degrees. The standard deviation reflects the interscan variance of this error. Significant improvements over the uncorrected approach and DEEP-DISORDER are denoted by * and $\dagger$, respectively (Wilcoxon signed-rank test, $p < 0.001$).

Volunteer Motion	Reconstruction Approach	Motion-Corrected Reconstruction Quality ($\mu \pm \sigma$)			
		$R = 1$, SSIM $\uparrow$	$R = 2$, SSIM $\uparrow$	$R = 1$, PSNR $\uparrow$	$R = 2$, PSNR $\uparrow$
Gradual	Uncorrected	0.929 ± 0.023	0.893 ± 0.027	30.7 ± 1.7	28.3 ± 1.5
	DEEP-DISORDER	0.940 ± 0.013	0.925 ± 0.026	31.2 ± 1.1	30.7 ± 1.7
	Deep-Reorder	0.951 ± 0.003	0.943 ± 0.010	31.8 ± 0.4	31.6 ± 0.9
Jittery	Uncorrected	0.857 ± 0.032	0.851 ± 0.039	27.7 ± 1.9	26.8 ± 1.8
	DEEP-DISORDER	0.947 ± 0.009	0.928 ± 0.018	32.1 ± 1.0	30.8 ± 1.5
	Deep-Reorder	0.955 ± 0.001	0.944 ± 0.005	32.6 ± 0.5	31.7 ± 0.8
Sweeping	Uncorrected	0.774 ± 0.005	0.789 ± 0.016	24.8 ± 0.7	24.8 ± 1.0
	DEEP-DISORDER	0.940 ± 0.005	0.928 ± 0.006	31.5 ± 0.5	30.5 ± 0.2
	Deep-Reorder	0.947 ± 0.004	0.936 ± 0.009	31.8 ± 0.5	31.0 ± 0.5

TABLE 3 Reconstruction quality on prospectively motion-corrupted data with intentional motion by volunteers.

both strategies, an additional variant was investigated that switches to random regrouping after a given number of iterations.

Sliding window regrouping groups adjacent shots with a stride of 1 between iterations. If our postulate holds that the original temporal correlation between adjacent shots is rapidly diminished by a few iterations of initial correction, then sliding window regrouping may be counterproductive because it exclusively and repeatedly groups adjacent shots.

The dilated strategy is motivated as follows. Since the first iteration always corrects in segments of multiple adjacent shots, the residual error between certain shots after original correction may retain some structure from the original motion trajectory. In the case of prolonged motion in the same direction, the three shots per original segment will thus retain a trend in the same direction. If the same motion happens for multiple segments, then

shots at similar relative positions within those segments will retain similar residual errors after initial correction. To investigate whether this can be exploited, the dilated strategy groups shots at similar relative positions during the second iteration. In technical terms, it samples with a dilation (sampling interval) of three shots, where three equals the segment size. One variant uses random regrouping in subsequent iterations, while the other variant continues using a dilation that increases by two per iteration.

Figure 8 compares the convergence of the various regrouping strategies. The sliding window approach converges substantially slower than random regrouping, except initially for nodding motion, which we discuss in the Discussion section. The dilated strategy converges modestly faster than random regrouping for the smooth Perlin motion pattern, where continuous motion

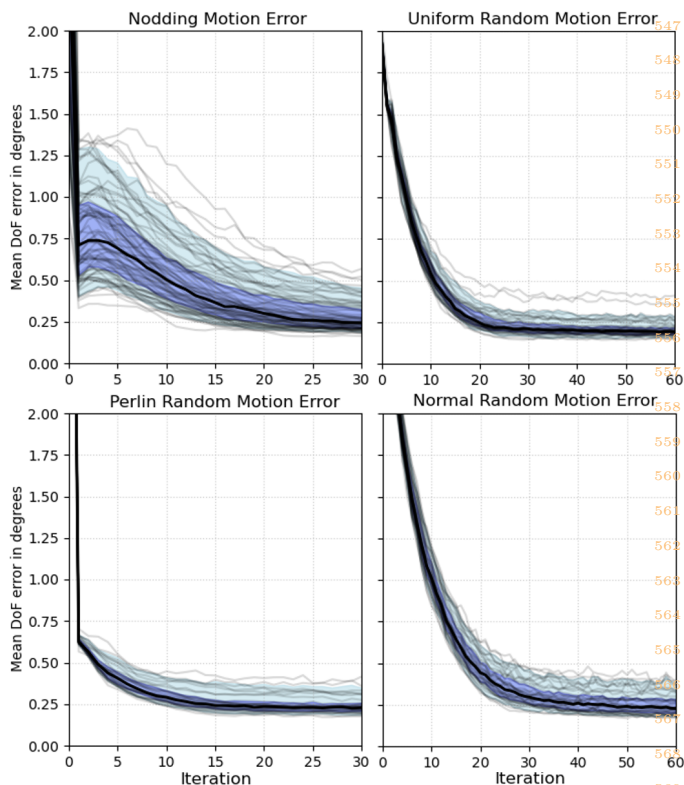


FIGURE 4 Mean rotation error per degree of freedom over iterations between ground-truth motion patterns and estimations by Deep-Reorder for retrospectively corrupted data. Zero iterations corresponds to no correction while one iteration corresponds to the previously proposed approach DEEP-DISORDER.

occurs in the same direction over a prolonged period, but slightly slower for the other motion patterns.

4 | DISCUSSION

We presented Deep-Reorder, a framework for motion correction in 3D MRI that enables motion estimation with an increased number of estimated motion states. The approach considerably increased image quality in cases with severe motion, which previously could not be corrected due to the limited temporal resolution.

A previously proposed approach DEEP-DISORDER only has enough information when it estimates motion between segments of 3 shots (1/44th of the data), and thus is less effective when motion occurs within those segments. The proposed Deep-Reorder approach on the other hand, gradually estimated a unique set of rigid motion parameters for each shot (1/132th of the data), enabling individual motion-correction for each, as demonstrated on retrospectively motion-corrupted 3D T1w TFE scans with a random motion state per shot. On retrospectively motion-corrupted scans with nodding

and shaking motion from a previous study applied to each shot, this led to an improvement in accuracy of estimated motion parameters from 2.59° , via 0.76° with DEEP-DISORDER, to 0.27° with Deep-Reorder. The reconstruction quality improved from 0.898 SSIM when uncorrected, via 0.973 with DEEP-DISORDER, to 0.992 SSIM with Deep-Reorder.

On prospectively motion-corrupted scans with intentional volunteer motion, we observed two situations. In some cases, often with light motion, the previously proposed approach DEEP-DISORDER sufficiently corrected for motion artifacts, in which case the scans could naturally not be further corrected. In the remaining cases where motion artifacts remained after correction by DEEP-DISORDER, the proposed Deep-Reorder considerably improved image quality, producing sharper and less noisy images. For scans corrupted with jittery motion, DEEP-DISORDER increased the SSIM from 0.857 to 0.947, while the proposed Deep-Reorder approach further increased it to 0.955 SSIM. Scans corrupted with gradual and sweeping motion patterns displayed a similar trend. After Deep-Reorder removed the majority of motion artifacts, a cold-diffusion-based refiner model provided a further visual reduction in noise and tiny motion artifacts that remained in the motion-corrected CG-SENSE reconstructions, although this was not reflected in a statistically significant quantitative improvement.

In the end, the proposed approach yielded robust correction across most cases, though a fraction of highly motion-degraded scans retained moderate noise-like residual artifacts. Future work could examine whether intrashot motion estimation can increase the temporal resolution of Deep-Reorder to further correct these cases, or investigate whether the method could be applied to sequences that do not have an inherent shot-like structure, such as gradient echo sequences. Alternatively, outlier rejection could be used to discard insufficiently corrected shots. For intrashot correction, if the underlying motion estimation relies on DISORDER sampling, attention should be paid to grouping sub-shots only in such a manner that a balanced distribution of frequencies is ensured for each segment. Additionally, an increase in temporal resolution will correspond to an increase in computational cost. The runtime of the underlying motion estimation algorithm could be a limitation for the proposed Deep-Reorder framework. Using DEEP-DISORDER as a motion estimation module, one iteration took about 8 minutes for $R = 2$. Thus, while an initial motion-corrected image is visible after this time, a thorough correction of intrasegment motion would take considerably longer depending on the severity of the motion and required number of iterations.

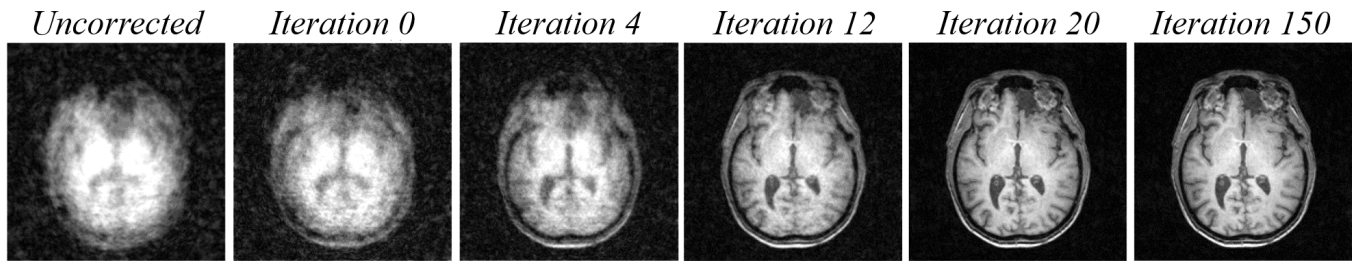


FIGURE 5 Correction quality across iterations on retrospectively motion-corrupted data with random motion. This example converged around 20 iterations, after which running more iterations had no meaningful effect.

This time could be substantially reduced by switching the registration component in DEEP-DISORDER to GPU-based registration, potentially using deep learning^{37,38,39}. Another option would be to combine the proposed iterative regrouping strategy with a neural network that directly estimates the motion parameters for each k-space segment, which would also be interesting for investigating the generalizability of the iterative regrouping approach with respect to different motion estimation components. A limitation of the evaluation may be that the sweeping motion pattern was performed in jittery steps. Consequently, large-amplitude gradual motion was only investigated in the retrospectively corrupted scans.

For the scans retrospectively corrupted with nodding motion, the error temporarily increased after initial correction. We speculate this is a result of the nodding pattern combining long periods of low motion with sharp motion peaks. These peak shots are most likely to be regrouped with low motion shots in resting position, of which there are many. This will improve the parameters of the peak shots but deteriorate the resting shots, thus not improving the motion error. In fact, since the motion states in such a segment are so different, the estimation inaccuracy may deteriorate the motion error. While not reflected in the metrics, the scan is hereafter easier to correct since the peaks are lower. By regrouping locally, the sliding window approach keeps the low motion shots in a good state, but consequently underperforms in the long run by insufficiently addressing the outlying peak shots.

We used random regrouping since it makes no heuristic assumptions and is not vulnerable to pathological looping. It was demonstrated that faster converging regrouping strategies can be designed if the motion patterns follow a certain distribution, such as a regrouping strategy which assumes that motion occurs in a similar direction for multiple shots. This is probably not the case in practice, where motion may often be abrupt. Moreover, the original correlation in time between shots is quickly diminished by motion correction, so predefined strategies only remain effective for a few iterations.

We suspect that more effective regrouping strategies may be possible, but that these should be informed by criteria derived from the image data or motion trajectories estimated so far. Investigating whether such strategies can reliably group shots that are closest in their current motion state is a challenging problem, and warrants investigation in future work. As another topic for future work, instead of DISORDER sampling, which we used to ensure a balanced distribution of low and high frequencies in each shot, the approach could be implemented using other compressed sensing incoherent sampling patterns such as CASPR with golden angle profile ordering⁴⁰. A final direction could be to investigate whether the approach could enhance the temporal resolution for non-rigid motion correction as well, a substantially more challenging task for which many more motion parameters need to be estimated.

In conclusion, the iterative regrouping approach Deep-Reorder introduced in this work improved the temporal resolution of an existing motion-correction approach called DEEP-DISORDER, enabling individual motion-correction of all 132 shots in 3D T1w MRI.

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DATA AVAILABILITY STATEMENT

The authors do not have permission to share data.

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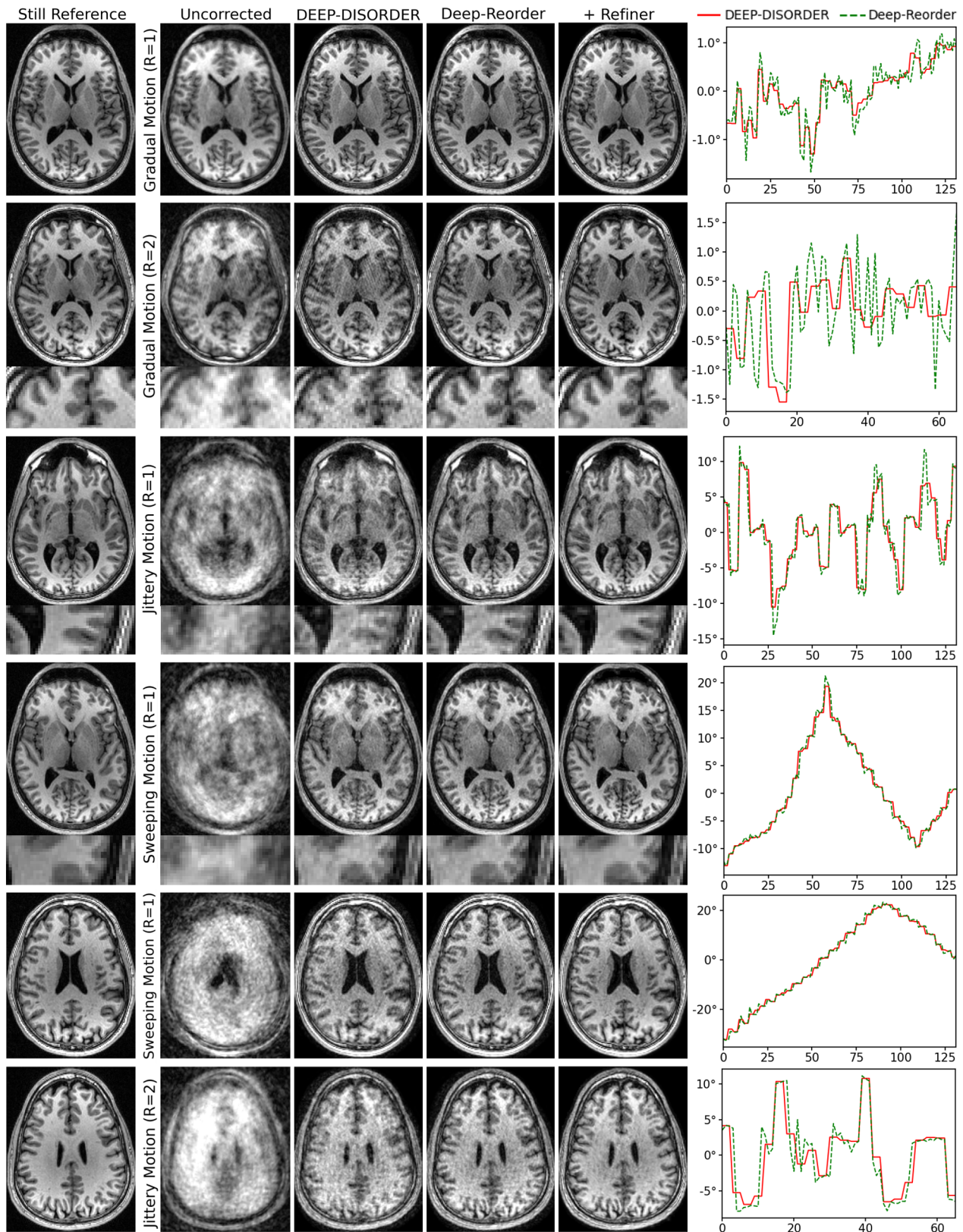


FIGURE 6 Reconstructions of motion-corrupted 3D MRI scans from volunteers performing intentional motion. From top to bottom, examples with increasingly more severe corruption are shown. The rightmost column of images shows the Deep-Reorder reconstructions after applying the refiner introduced in Section 3.3. The zoomed-in regions exemplify differences that may be difficult to discern at full scale. Finally, the estimated rotational motion about the z-axis (feet-head) is displayed.

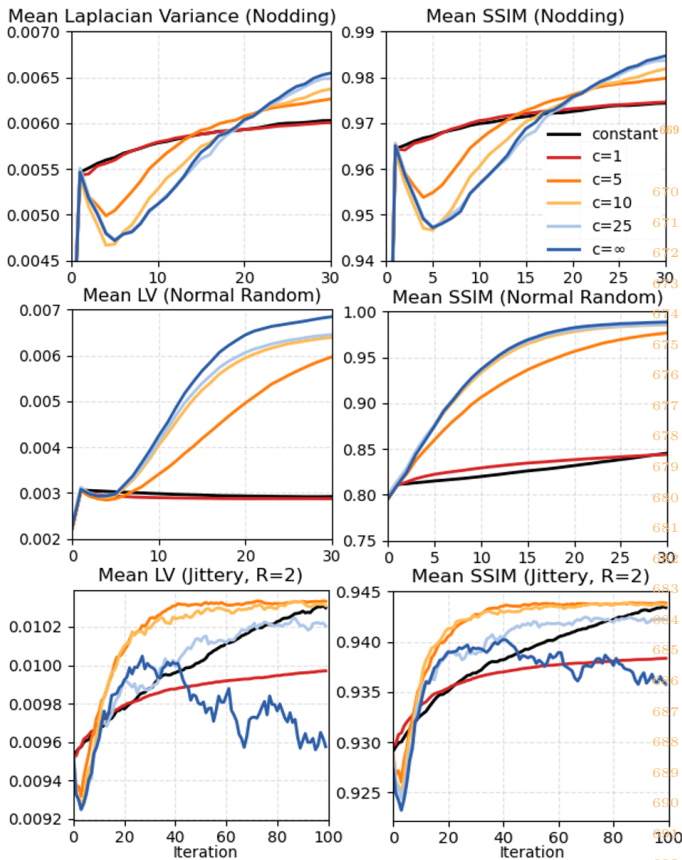


FIGURE 7 Parameter study investigating the choice of the scale factor c used in Equation 2, or using a constant limit of $\Delta_{max} = 0.1$. Lower values of c correspond to a stronger motion update limit, while higher values correspond to a weaker limit, and ∞ corresponds to no limit. The top and middle rows present 15 retrospectively corrupted scans while the bottom row presents 3 prospectively corrupted scans.

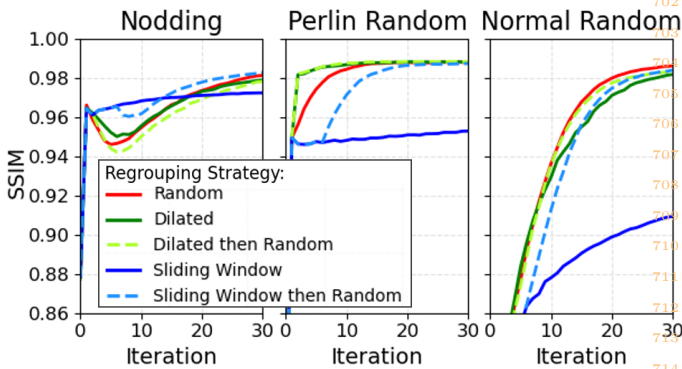


FIGURE 8 Mean SSIM of alternative regrouping strategies for different motion patterns. The dashed blue line represents sliding window switched to random regrouping after five iterations, while the dashed green line represents dilated switched to random regrouping after the second iteration.

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